# A Proof Dependency Manifest for a Candidate Computer-Assisted Proof of $f(7,7) = 38$

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14 September 2026

**Abstract**

This document is a dependency manifest, not a paper. It states exactly what the candidate proof of $f(7,7) = 38$ consists of, and separates published external mathematics from project-original arguments, computational reductions, external database checks, certificate verification, Lean-kernel verification, and the items still requiring human review. Every claim carries a source and a verification status. It is written so that a referee can locate the weakest link quickly.

**Status.** This document describes a *candidate* computer-assisted proof. All currently known computational components have been independently checked or formally verified, but the interpretation of AKR Observation 6 / Lemma 7 remains subject to external human review. $f(7,7) = 38$ **is not a Lean theorem**: the strongest formalised statement still carries two hypotheses (§1.1). The result should not be treated as established.

## Disclosure of generative-AI use

Generative-AI systems — principally Anthropic's Claude, together with OpenAI's ChatGPT — were used substantially throughout this project. Their use was not incidental: it extended to proof exploration and the search for the reduction strategy; literature-guided reasoning about the published results; generation of essentially all software, in Python, C++ and Lean; the computational experimentation itself; debugging; the construction of redundant and independent verification paths; the Lean formalisation; source auditing of the cited papers; and the drafting of the project's documentation, including this manifest.

The named author is responsible for the claims made in this document. AI-generated output was not treated as inherently trustworthy at any point. Each major claim is discharged by an explicit, separately recorded mechanism — source audit against the published papers, independent reimplementation in a second language, an externally sourced dataset, a machine-checkable certificate with a standalone checker, or Lean-kernel verification — and §3 records which mechanism applies to which claim. Several substantive errors introduced during AI-assisted development were found by those mechanisms and are catalogued in §11; they are retained there rather than quietly repaired, because they calibrate how much checking the argument has actually received.

Project artifacts and provenance — source code, generated data, certificates, build and verification logs, the audit documents, and the complete development transcripts — are preserved in the project repository.

No claim is made here about conformance with any particular publisher's or institution's policy on generative-AI use; that must be assessed separately against the policy in force.

# 1 Claim and current status

Let $f(m, n)$ denote the maximum number of boundary intersection points of two simple polygons with $m$ and $n$ sides in general position.

**Theorem 1** (Claimed)**.** $f(7, 7) = 38$.

The problem is due to [6], who asked for $f(m, n)$ in general and settled the even cases. The odd–odd cases have resisted; a 2012 preprint claiming $f(n, n)$ for odd $n$ [7] was withdrawn by its author as faulty and is not relied on here.

Three facts frame the problem.

1. 38 **is realizable.** An explicit pair of integer-coordinate heptagons attains 38 crossings (§9, LB38); this is verified by the Lean kernel.

2. **The published state of the art leaves only** 40**.** Černý et al. [5, Theorem 4] prove $f(k, \ell) \le k\ell - \lceil k/6 \rceil - \ell$ for odd $7 \le k \le \ell$, which at $k = \ell = 7$ gives $f(7, 7) \le 40$; [2] is asymptotically optimal but vacuous at $m = n = 7$. Crossings of two closed curves in general position are even, so $f(7, 7) \in \{38, 40\}$ and the whole question is whether 40 is attainable.

   *This bound is context, not a dependency.* Importing it would shorten the argument — $I \le 40$, $I \ne 40$ and parity would give $I \le 38$ at once, with no $I = 42$ case — but [5, Theorem 4] rests on [5, Lemma 3], a four-page case analysis this project has not audited, and importing it would contradict the project's own accounting of its dependencies. The argument here instead uses only [5, Lemma 1] (C1), which gives the weaker cap $I \le 42$, and excludes $I = 42$ separately in §5.4. Černý Theorem 4 is therefore independent corroboration of the upper bound, not an input to it.

3. **The upper bound is a proof by contradiction.** One assumes a pair with $I = 40$ and derives a contradiction. Everything in §2–§8 serves that single reductio.

We do *not* claim Theorem 1 is established. We claim that no computational gap is currently known, and that the remaining risk is concentrated in one identified place (A7-EXT, §4.5).

## 1.1 What is a Lean theorem today, and what is not

The Lean development (§9) is large enough that a reader can mistake its coverage for coverage of Theorem 1. It is not. The strongest statement it contains is

```
theorem IntersectionParity.isGreatest_38_final
    (hclass : forall K : Config, exists mcs MP, MP.card = 5 /\ ... ) --  X1 + X2
    (hne40  : forall K : Config, crossCountC K <> 40)                --  UB40
    : IsGreatest MainTheorem.attained 38
```

and it still carries those *two hypotheses*. Neither is proved anywhere in the development:

- `hclass` is the exceptional-class invariant together with the order-type crossing fact — X1 and X2 of §9, i.e. EX and E4 — asserted for *every* configuration;
- `hne40` is $I \ne 40$, that is the entire 13,003,620-graph enumeration, UB40. It is **not** reachable by the Lean kernel on this architecture; five routes were tried and all are dead (§9).

Two further gaps sit outside the theorem statement and are not visible in it. `Config` is an *orientation-level* object over `Pt` $:= \mathbb{Z} \times \mathbb{Z}$: "simple" means a conjunction of integer orientation predicates, not "the closed polygonal curve is a Jordan curve". And the reduction from arbitrary position to general position is GP, AKR §9.1, which is cited and not formalised. So even with both hypotheses discharged, the theorem would read "38 is the maximum crossing count over integer-coordinate, orientation-level heptagon configurations", and two further external steps would be needed to reach $f(7, 7) = 38$.

What *is* unconditional, for an arbitrary `Config`, is the following, and it is worth stating because it is a genuine bracket:

$$I + |E^D| = 49, \qquad \delta(G^D) \geq 1, \qquad I \leq 42, \qquad I \text{ even}, \qquad 38 \in \text{attained},$$

so the answer lies in $\{38, 40, 42\}$ with no hypotheses at all (`MainTheorem.attained_38_and_le_42`, `attained_mem_three`). Adding `hclass` removes 42 (`Exclude42.not_crossCount_42`). Only 40 resists.

## 2 One-page proof flow

Write $G^D$ for the bipartite *disjointness graph* on the $7+7$ sides, with $(p_i, q_j) \in E^D$ iff the closed segments $p_i$ and $q_j$ are disjoint. Since two non-collinear segments meet at most once, crossings biject with intersecting side pairs, so

$$I + |E^D| = 49.$$

The reductio proceeds as follows; the label under each arrow names the justification.

| | | |
|---|---|---|
| $I = 40$ | $\xRightarrow{I + \lvert E^D\rvert = 49}$ | $\lvert E^D\rvert = 9$ |
| | $\xRightarrow{\mathsf{C1}\ (\text{Černý L1})}$ | $\delta(G^D) \geq 1$ |
| | $\xRightarrow{\mathsf{LA}\ (\text{Lemma A + parity})}$ | $P$ and $Q$ are both *exceptional* |
| | $\xRightarrow{\mathsf{OT},\mathsf{EX}\ (\text{order-type classification})}$ | $(\text{profile}_P, \text{profile}_Q) \in \{14\} \times \{14\}$ |
| | $\xRightarrow{\mathsf{GR},\mathsf{COMP}\ (\text{enumeration; } \mathsf{A5\text{-}C},\mathsf{A7\text{-}EXT},\mathsf{C2})}$ | 225 terminal triples on 121 graphs |
| | $\xRightarrow{\mathsf{A5\text{-}A},\mathsf{CERT},\mathsf{LEAN\text{-}END}}$ | every terminal triple violates AKR Lemma 5 |
| | $\Longrightarrow$ | $\bot$. |

Hence $I \neq 40$; with $I \leq 42$ (C1), $I \neq 42$ (§5.4) and $I$ even, this gives $I \leq 38$, and the explicit construction LB38 attains 38. Therefore $f(7, 7) = 38$.

*Remark.* All of this is conditional on the general-position regime GP, imported from [2, §9.1]; see §4.1.

## 3 Dependency table

| ID | Claim | Type | Source | Verification | Remaining risk |
|---|---|---|---|---|---|
| GP | General-position regime may be assumed w.l.o.g. | published theorem | [2, §9.1,§3] | SOURCE-AUDITED | none known; project's own earlier proof was wrong and is removed |
| C1 | A line meets a simple 7-gon in at most 6 sides; hence $\delta(G^D) \geq 1$ | published theorem | [5, Lemma 1] | SOURCE-AUDITED | none known |
| C2 | Forbidden local matching on 4 consecutive sides with $D \in \text{int} \triangle ABC$ | published theorem | [5, Lemma 2] | SOURCE-AUDITED | none known |

| ID | Claim | Type | Source | Verification | Remaining risk |
|---|---|---|---|---|---|
| C2-BRIDGE | The implemented filter predicate fires iff an *exact induced* perfect matching exists | project inference | this project | LEAN-KERNEL-CHECKED | certifies the Lean transcription of the filter, not the Python/C++ source |
| A5-C | Lemma 5 applied to *consecutive* side pairs in different components | published theorem | [2, Lemma 5] | SOURCE-AUDITED | none known |
| A5-A | Lemma 5 applied to *avoiding* side pairs | published theorem | [2, Lemma 5] | SOURCE-AUDITED; bridge LEAN-KERNEL-CHECKED | none known |
| A7-EXT | Under the intended reading of "associated"/"is hooking", AKR Lemma 7 gives: adjacent $H$-vertices have disjoint status sets | published theorem + interpretation | [2, Lemma 7], [1] | VERY STRONGLY CORROBORATED | **principal open item**; Observation 6 is unproved and *false as printed* (§11, item 6), though the failure cannot reach $S(e)$; four bridges from the machine checks to the geometry remain prose |
| A7-BRIDGE | Non-empty status sets + disjoint status sets across edges $\Rightarrow$ $H$ bipartite | project inference | this project | LEAN-KERNEL-CHECKED | none known; conditional on A7-EXT |
| LA | Lemma A: an uncrossable 4-set forces $\lvert E^D\rvert \geq 10$, parity gives $\geq 11$, so $I \leq 38$ | project lemma | this project | LEAN-KERNEL-CHECKED (counting core) | $h_2$ is now *derived* (LEAN-GEO); $h_1$ and $(\star)$ remain hypotheses |
| PAR | $I$ is even for every configuration in general position | project lemma | this project, §5.3 | LEAN-KERNEL-CHECKED | none; proved outright 2026-09-14 |
| OT | There are exactly 135 order types of 7 points | external dataset cross-check | [4, 3] | EXTERNALLY CROSS-CHECKED (exact set equality) | none known |
| EX | 6,342 simple heptagons; 16 exceptional; 14 profile classes | finite enumeration | this project | INDEPENDENTLY REIMPLEMENTED; 6,342 EXTERNALLY CROSS-CHECKED | none known |
| GR | Exactly 13,003,620 candidate disjointness graphs | finite enumeration | this project | INDEPENDENTLY REIMPLEMENTED + closed form | none known |
| COMP | Filter pipeline down to 121 graphs / 225 triples | finite enumeration | this project | INDEPENDENTLY REIMPLEMENTED (exact set match) | inherits A7-EXT |
| CERT | Each of the 225 triples violates the AKR-L5 consequence | certificate | this project | CERTIFICATE-CHECKED | completeness of the list is not certified |

| ID | Claim | Type | Source | Verification | Remaining risk |
|---|---|---|---|---|---|
| LEAN-LB | The explicit pair has 38 crossings and is simple | Lean formalization | `lean/LowerBound.lean` | LEAN-KERNEL-CHECKED | simplicity is orientation-level, not Jordan-curve |
| LEAN-LA | Counting and parity core of Lemma A | Lean formalization | `lean/LemmaA.lean` | LEAN-KERNEL-CHECKED | proved as an implication |
| LEAN-END | All 225 cases fail the AKR-L5 condition; conclusion $\leftrightarrow$ predicate | Lean formalization | `lean/Endgame.lean` | LEAN-KERNEL-CHECKED | completeness of the 225 list not formalized |
| LEAN-GEO | A closed polygon meets a line evenly; hence $\delta(G^D) \geq 1$ and Lemma A's $h_2$ | Lean formalization | `lean/SegmentGeometry.lean` | LEAN-KERNEL-CHECKED | Mathlib-free; orientation level |
| LEAN-I2 | The stored mask bit means point-set disjointness of the two closed sides in $\mathbb{R}^2$ | Lean formalization | `DisjointnessGraph` + `PointSetSemantics` | LEAN-KERNEL-CHECKED | I2a+I2b; uses Mathlib and $\mathbb{R}$, so only this step leaves the Mathlib-free base |
| LEAN-MAIN | 38 is the greatest attained crossing count over `Config` | Lean formalization | `MainTheorem`, `IntersectionParity` | proved *from* `hclass` and `hne40` | **not a proof of** $f(7,7) = 38$; see §1.1 |
| LB38 | $f(7,7) \geq 38$ | construction | this project | LEAN-KERNEL-CHECKED | orientation-level simplicity; no Jordan-curve step |
| UB42 | $I = 42$ is impossible | project inference | this project, §5.4 | LEAN-KERNEL-CHECKED given X1,X2 (`Exclude42`); also exhaustively swept | consumes C1, LA, EX only — *not* A7-EXT; C1 gives only $I \leq 42$, so this branch is needed |
| UB40 | $I = 40$ is impossible, hence $f(7,7) \leq 38$ | composite | GP, C1, C2, A5-C, A5-A, A7-EXT, A7-BRIDGE, LA, COMP | NOT FORMALIZED; conditional on A7-EXT | **fails if A7-EXT fails**. Carried in Lean as the bare hypothesis `hne40`, which is *out of reach* of the kernel (§9) |

# 4 External mathematical dependencies

## 4.1 GP — general position, AKR §9

**Role.** The whole combinatorial argument is carried out in a general-position regime. GP is what licenses that restriction.

**Source.** [2, §9.1, and §3]. Their perturbation removes the relevant degeneracies without decreasing the number of boundary intersections.

**Exact consequences used, and nothing stronger.**

(GP1) both polygons remain simple;

(GP2) no vertex of one polygon lies on the boundary of the other, so every intersection point is interior to two sides — this is what allows intersections to be counted as proper crossings;

(GP3) no three vertices of $P$ and $Q$ combined are collinear;

(GP4) no two sides of $P$ and $Q$ combined are parallel;

(GP5) consequently the supporting line $\ell(q_j)$ of any side of $Q$ contains no vertex of $P$ (it passes through two vertices of $Q$, so a vertex of $P$ on it would violate (GP3)). This is exactly the hypothesis that Lemma A needs, and it is the *only* kind of line at which Lemma A is ever instantiated.

*Remark* (A superseded project argument). Earlier versions of this project proved GP by a project-specific perturbation rule. That argument is **incorrect**: its rule for a vertex-on-vertex degeneracy, "push the coincident vertex into $\mathrm{Cone}_Q(w)$", destroys the intersection point whenever $\mathrm{Cone}_P(v) \subset \mathrm{Cone}_Q(w)$, yielding 0 crossings instead of $\geq 1$. It also never addressed the case in which one side participates in several degeneracies at once. It has been removed from the active proof and replaced by the citation above. See §11.

*Remark* (Counting model). [2, §9.3] explicitly leaves the variant counting intersecting *side pairs* rather than intersection *points* as future research. Everything here is in the intersection-*point* model, and the result must not be restated in the pairs model.

### 4.2 C1 — Černý Lemma 1

A line meets a simple $k$-gon an even number of times, hence at most $k-1$ times for odd $k$ [5, Lemma 1]. Applied to $\ell(p_i)$ against $Q$ and to $\ell(q_j)$ against $P$, every side misses at least one side of the other polygon. The single consequence used is

$$\delta(G^D) \;\geq\; 1.$$

Together with $|E^D| = 49 - I$ this also gives $I \leq 42$.

### 4.3 C2 — Černý Lemma 2

The only consequence used is a local forbidden configuration. Let $A, B, C, D, E$ be five consecutive vertices of $P$, giving four consecutive sides $AB, BC, CD, DE$, and suppose $D \in \operatorname{int} \triangle ABC$. Then for no four sides of $Q$ is the induced subgraph of $G^D$ on those four sides of $P$ and those four sides of $Q$ *exactly* a perfect matching [5, Lemma 2].

The geometric hypothesis depends only on the order type and is tabulated per exceptional class as the *Lemma-2 windows*. The word *exactly* is load-bearing: a block containing a perfect matching *plus* an extra edge is not the forbidden pattern *for that block*. (It does not follow that the graph survives the filter — another block may still trigger it.)

### 4.4 A5 — AKR Lemma 5

**Statement used.** If two sides $p_a, p_b$ of $P$ that are *consecutive or avoiding* lie in different components of $G^D$, then there are consecutive sides $q_i, q_{i\pm1}$ of $Q$ with

$$(p_a, q_i),\ (p_b, q_{i\pm1}) \in E^D \qquad \text{and} \qquad (p_a, q_{i\pm1}),\ (p_b, q_i) \notin E^D.$$

**A5-C, consecutive use.** Applied to every consecutive pair in different components; this is the first filter of the pipeline.

**A5-A, avoiding use.** Two non-adjacent sides are *avoiding* when neither contains the intersection point of their supporting lines. This is an order-type predicate, tabulated per exceptional class. A5-A is the final filter and the one the certificate records.

**Combinatorial consequence.** Writing $N(\cdot)$ for neighbourhoods in $G^D$, the displayed conclusion is equivalent to

$$\exists\, k \in N(p_a) \setminus N(p_b), \quad \exists\, k' \in N(p_b) \setminus N(p_a), \qquad k' \equiv k \pm 1 \pmod 7. \tag{1}$$

Both directions are immediate by unfolding: $k \in N(p_a) \setminus N(p_b)$ says exactly $(p_a, q_k) \in E^D$ and $(p_b, q_k) \notin E^D$.

*Remark.* This equivalence is **Lean-kernel-checked**, as `akr5Conclusion_iff_adjacent` in `lean/EndgameDefs.lean`, and depends on no axioms. It matters because it shows the implemented predicate is neither a weakening nor a strengthening of the published conclusion.

*Remark.* If $\mathrm{CC}(p_a) \neq \mathrm{CC}(p_b)$ then $N(p_a) \cap N(p_b) = \emptyset$, since a shared neighbour would join them in one component. The set differences in (1) are then the neighbourhoods themselves.

### 4.5 A7-EXT and A7-BRIDGE — AKR Lemma 7 and bipartiteness of $H$

This dependency was previously recorded as a single item. It bundles two things of very different character, and is split here:

$$\underbrace{\text{A7-EXT}}_{\text{published theorem+interpretation}} \implies \underbrace{\text{A7-BRIDGE}}_{\text{project inference, Lean-checked}} \implies H \text{ bipartite.}$$

**Definition of $H$.** Fix $P$ (symmetrically for $Q$). Vertices are the relevant consecutive pairs, $V(H) = \{\, i : \mathrm{CC}(p_i) \neq \mathrm{CC}(p_{i+1}) \,\}$, and $i \sim_H k$ iff the two pairs share no side, i.e. $(k-i) \bmod 7 \notin \{0,1,6\}$, *and* the four sides $p_i, p_{i+1}, p_k, p_{k+1}$ lie in four distinct components of $G^D$. Condition (b) is imposed directly, so an $H$-edge reproduces Lemma 7's hypothesis exactly.

For $e \in V(H)$ let $A(e)$ be the set of pairs of consecutive sides of $Q$ associated to $e$ *in the narrow sense* — carrying, beyond the two edges of $E^D$, the *cone dichotomy* of Lemma 5 — and put

$$S(e) = \{\text{hooking} : \exists \alpha \in A(e),\ I(\alpha) \in \mathrm{Cone}(e)\} \cup \{\text{hooked} : \exists \alpha \in A(e),\ I(e) \in \mathrm{Cone}(\alpha)\}.$$

$S(e) \neq \emptyset$: $e \in V(H)$ is Lemma 5's hypothesis, so $A(e) \neq \emptyset$, and Lemma 5's dichotomy realises at least one status.

*Remark* (Which extra conditions the narrow sense actually needs). Lemma 5 delivers, beyond the two edges $(p_i, q_j), (p_{i+1}, q_{j\pm1}) \in E^D$, both the two cross non-edges and the cone dichotomy. Only the second is a genuine strengthening.

*The cross non-edges are automatic.* If $(p_i, q_{j\pm1}) \in E^D$ then $p_i - q_{j\pm1} - p_{i+1}$ is a path in $G^D$, so $\mathrm{CC}(p_i) = \mathrm{CC}(p_{i+1})$, contradicting $e \in V(H)$. This is pure connectivity; bipartiteness of $G^D$ plays no part. Checked exhaustively over all $2^{16}$ bipartite graphs on $4+4$ vertices: 110,304 qualifying configurations, 0 violations.

*The cone dichotomy is not.* Adding the non-edges does *not* recover Observation 6: for $a = [(0,0),(2,0)]$, $b = [(0,0),(0,3)]$, $A = [(-1,10),(\frac{1}{10},\frac{6}{5})]$, $B = [(-1,10),(\frac{7}{5},-2)]$ all four edge and non-edge conditions hold with $a, b$ consecutive, yet the axis $y = -10x$ leaves both $a$ and $b$ in $y \geq -10x$.

So the narrow notion used here is **edges + cone dichotomy**, the non-edges being free. A formalisation that instead defined it as "edges + cross non-edges" and then asserted Observation 6 would be unsound.

#### A7-EXT — external, very strongly corroborated

> Under the intended interpretation of "associated" and "is hooking", [2, Lemma 7] yields: if $e \sim_H e'$ then $S(e) \cap S(e') = \emptyset$.

This is the one item that remains subject to external human review, for three reasons, in increasing order of comfort.

1. AKR's Observation 6 (the Axis Property) is stated *without proof* in both versions, and is *false as printed*. The counterexample is not a bare four-segment picture but a pair of genuine simple polygons in full general position,

$$P = (2,0),(3,4),(-2,0); \qquad Q = (4,-4),(-3,1),(0,-1),(3,3),(2,1),$$

verified in exact integer arithmetic (§11, item 6). Here $q_0, q_4$ is associated to $p_0, p_1$ under the printed definition — indeed the cross non-edges hold too — yet the axis separates nothing. Observation 6 is used throughout Lemma 7's case analysis.

This does *not* refute A7-EXT. Each status is by definition one disjunct of the cone dichotomy, so an associated pair failing the dichotomy contributes nothing to $S(e)$ and can never participate in the hypothesis. Equivalently, Lemma 7's hypothesis supplies the dichotomy exactly on the pairs that matter (Remark 4.5).

2. The conference version [1] opens its proof of Lemma 7 by *selecting* the associated pairs subject to the cone conditions $I(q_{i'}, q_{i'\pm1}) \in \mathrm{Cone}(p_i, p_{i+1})$, which forces the existential reading of "is hooking" used above. The journal version deleted those clauses, leaving its own next sentence unsupported. This strongly indicates the intended reading but does not state it in one place.

3. [2, §10.2], journal only, reports the authors' own independent computer verification of Lemma 7: 158,830 arrangements of eight lines, 840 assignments each after symmetry, yielding 14 configurations of which 13 fail Lemma 5's hooking/hooked pattern. This corroborates the *statement*. It also corroborates the *interpretation*: the stated ground for discarding those 13 is that "some associated pair did not conform to the hooking/hooked pattern of Lemma 5", a sentence that is coherent only if "associated" already carries the dichotomy.

   Two of the four published numbers in §10.2 do not reproduce, neither affecting the conclusion. 158,830 is a *nine*-element count (abstract order types on 9 points); $x$-monotone PSLAs of 8 pseudolines correspond to abstract order types on 9 points, the ninth element being the line at infinity, so a complete but redundant superset was enumerated — harmless for an impossibility claim. $840 = 8!/48$ quotients by Aut of the disjointness graph alone ($\cong Q_3$, order 48), a group that does not preserve which pairs are consecutive; the stabiliser of the configuration has order 16, giving $8!/16 = 2520$, reproduced by brute force over $S_8$. Both are fair questions for the authors.

### Independent verification of A7-EXT

An oriented-matroid (rank-3 chirotope) encoding over the twelve segment endpoints makes the both-hooking and both-hooked patterns UNSAT, while the mixed patterns of AKR's Figure 6 are SAT. Unsatisfiability over a *superset* of realisable configurations is sound for an impossibility claim, so under-constraining is safe and only over-constraining would be fatal. The encoding was attacked on exactly that point:

- every encoded predicate was checked against the paper, and each constraint is a *necessary* consequence of the hypothesis;
- **positive control:** AKR's own Figure 6 configuration, reconstructed from integer line coefficients in exact rationals, satisfies all 64,566 clauses — so the encoding is not vacuously over-constrained;
- an independent re-encoding from scratch reproduces the verdicts;
- re-solved with CaDiCaL, Glucose and MiniSat, all agreeing;
- DRAT refutations verified by a from-scratch RUP/RAT checker, itself validated on PHP(4) and PHP(5);
- the chirotope axiom set validated by reproducing the published enumeration counts $1, 1, 1, 4$ for $n = 3, \ldots, 6$ (OEIS A006248);
- **case exhaustion, after a repair.** The original coverage argument reduced the adjacency cases by a symmetry orbit and was *wrong* (§11, item 7); the PQ-adj verdicts it produced were vacuous. It is superseded by a derivation that uses no orbit argument at all: in exact $\mathbb{Z}_7$ arithmetic the coincidence equations give five types per polygon, hence $5 \times 5 = 25$ combined, confirmed by an exhaustive sweep of 50,176 labelled cases with 0 violations. All 25 are

then solved separately, with both-hooking and both-hooked UNSAT in every one. That is 125 instances and 125 certificates — 110 DRAT refutations and 15 explicit models — with no solver-only verdict, each certificate bound to its formula by SHA-256 and re-derived from source before checking.

Separately, a direct search for a counterexample on real heptagon pairs over exact integers, with simplicity and full general position enforced, examined ∼57,000 admissible configurations and 306 associated pairs on consecutive pairs in different components: 0 violations.

*Remark* (Why this is not VERIFIED INDEPENDENTLY). The machine checks are certificates about a chirotope encoding. Four steps connecting them to the geometric statement are hand-checked prose and none is certified: (i) general position of the $\leq 12$ vertices is WLOG (inherited from GP; perturbation can change $G^D$); (ii) "four distinct components" yields exactly the encoded pair pattern — 24 requirements (16 cross-polygon, 4 within $P$, 4 within $Q$), of which those sharing an endpoint carry no expressible constraint; (iii) the enumerated case list exhausts $n = 7$ — now derived exhaustively in exact arithmetic rather than by the orbit argument that turned out to be wrong, but by a project script, not by a certificate; (iv) Lemma 7's quantification matches the existential $S(e)$. The translation of "segments cross" and "point in cone" into chirotope conditions is validated by exact random testing, not by proof. And $S(e) \neq \emptyset$, needed alongside A7-EXT, rests on Lemma 5, which is machine-checked nowhere.

**A7-BRIDGE — project inference, lean-kernel-checked**

> If every relevant vertex has a non-empty status set, and adjacent vertices have disjoint status sets, then $H$ is bipartite.

*Proof.* Let $e \sim_H e'$. Both $S(e), S(e')$ are non-empty and disjoint subsets of a two-element set, hence distinct singletons. So if $S(e) = \{\text{hooking}, \text{hooked}\}$ then $e$ has no $H$-neighbour at all; every non-isolated vertex has $|S(e)| = 1$; every $H$-edge joins opposite singletons; and $e \mapsto S(e)$ is a proper 2-colouring of the non-isolated part, isolated vertices being irrelevant to 2-colourability. □

This inference is formalised in `lean/Lemma7Bridge.lean` (§9). The disjointness condition is an explicit *hypothesis*; AKR Lemma 7 is not assumed, not axiomatised, and not named in the file.

*Remark* (What the Lean check does and does not buy). It settles the "different associated pairs witness different statuses" worry completely: a both-status vertex is *proved* isolated, so the scenario in which Lemma 7 is applied with inconsistent choices on different $H$-edges is vacuous. It says nothing whatever about whether AKR's published language licenses A7-EXT. Lean verifies the project's implication; it does not verify AKR's geometric theorem.

# 5 Project-original mathematical reductions

Four items: Lemma A and the exceptional classification, which feed the enumeration; parity, which fixes the two branches to be excluded; and the exclusion of $I = 42$, which is one of them. Nothing here is imported.

## 5.1 Lemma A

**Lemma 2** (Lemma A)**.** *Let $S \subseteq \text{sides}(P)$ satisfy*

$$(\star) \qquad \forall \ell, \quad \ell \cap V(P) = \emptyset \implies \ell \text{ misses at least one member of } S.$$

*Then $|E^D| \geq n + (m - |S|)$.*

*Proof sketch.* For each side $q_j$ of $Q$, (GP5) gives $\ell(q_j) \cap V(P) = \emptyset$, so $(\star)$ applies and yields $p \in S$ with $\ell(q_j) \cap p = \emptyset$, hence $q_j \cap p = \emptyset$. This is $n$ edges with distinct $Q$-endpoints, all with $P$-endpoint in $S$. By C1 each $p \notin S$ misses some side of $Q$, giving $m - |S|$ further edges with distinct $P$-endpoints outside $S$. The two families are disjoint because their $P$-endpoints lie inside versus outside $S$. □

For $m = n = 7$ and $|S| = 4$ this gives $|E^D| \geq 10$; since $I$ is even, $|E^D| = 49 - I$ is odd, so $|E^D| \geq 11$ and $I \leq 38$. Call $P$ *exceptional* when no 4-subset of its sides satisfies $(\star)$. Then

$$I = 40 \implies P \text{ and } Q \text{ are both exceptional.}$$

**Lean boundary.** `lean/LemmaA.lean` kernel-checks the counting and the parity upgrade (`card_E_ge`, `card_E_ge_eleven`, `crossings_le_38`) as an *implication*. Of the two geometric premises:

- $h_2$ — that each $p \notin S$ is disjoint from some $q_j$ — is *derived*, for an arbitrary configuration, by the parity argument of `SegmentGeometry.lean` (LEAN-GEO). It is no longer an assumption;
- $h_1$ — that each $q_j$ is disjoint from some member of $S$ — remains an explicit hypothesis, discharged in Lean only for the one certificate pair.

The $(\star)$ condition itself is not formalized. Separately, `LemmaABridge.lean` connects Lemma A's `Finset` cardinality to the mask-level `degSum` used everywhere else, which is what lets the $|E^D| < 10 \Rightarrow$ "exceptional" step be applied rather than assumed (X3).

## 5.2 Exceptional classification

All predicates used — simplicity, crossing sets, the avoiding relation, the Lemma-2 windows — are Boolean functions of the order type of the seven vertices. The classification is therefore finite:

$$\begin{aligned} 135 \text{ order types } \times \text{ 360 cyclic orders } &\longrightarrow \text{ 6,342 simple heptagons} \\ &\longrightarrow \text{ 16 exceptional } \longrightarrow \text{ 14 classes.} \end{aligned}$$

The 6,342 split as 6,007 with an uncrossable 3-set, 319 with an uncrossable 4-set, and 16 exceptional. Each of the 14 classes has $|M| = 5$, exactly five maximal crossing sets all of size 6, and two avoiding pairs — except one class with none.

*Remark* (Strict versus loose, and why it matters). Two readings of "uncrossable" must be distinguished.

**strict:** $(\star)$ above versus **loose:** no line at all meets all four closed sides.

Measured over all 6,342 simple heptagons these give

$$\textbf{strict} \rightsquigarrow 16 \text{ exceptional}, \qquad \textbf{loose} \rightsquigarrow 2{,}268 \text{ exceptional},$$

differing on 2,252 objects. The *strict* reading is the correct one: Lemma A is only ever instantiated at lines $\ell(q_j)$, which avoid $V(P)$ by (GP5). The 2,252 differing objects are exactly those 4-sets whose only transversals pass through a vertex of $P$ — and no $\ell(q_j)$ can be such a line. Had the loose phrasing been implemented, the finite search would have been infeasible.

## 5.3 Parity: $I$ is even

**Role.** With $I \leq 42$ from C1, parity is what reduces the upper bound to the two branches $I = 42$ and $I = 40$. Without it, 39 and 41 would also need excluding.

**Status.** LEAN-KERNEL-CHECKED as of 2026-09-14 (`IntersectionParity.crossCountC_even`), for an *arbitrary* configuration rather than the certificate pair. It was previously carried as an assumption. The three standard axioms only.

**The argument.** Two closed curves in general position cross an even number of times. Formalised so as to stay inside $\mathbb{Z}$, introducing no reals and so preserving the Mathlib-free trust base of the geometry group:

1. an inside predicate for a point with respect to $\partial Q$, defined by *ray-crossing parity* under a sheared height order — the lexicographic $(y, x)$ order, which `hlt_eq_decide` identifies with the order of the integer functional $p \mapsto Np_2 + p_1$ once $N$ exceeds the $x$-spread, i.e. a ray obtained from the horizontal by an infinitesimal tilt;
2. a four-point bridge lemma relating the orientation predicates of the two segment endpoints to that parity;
3. the per-side lemma `crossRow_parity`: side $i$ of $P$ crosses $\partial Q$ an odd number of times exactly when its two endpoints differ in inside-ness;
4. the cyclic sum: each vertex indicator occurs in exactly two consecutive sides, so summing (3) around the closed polygon doubles every term. This is the same cancellation `SegmentGeometry.cycChanges_even` performs for a line.

*Remark* (Not the same as the line parity already present). `SegmentGeometry.metCount_even` — a closed polygon meets a *line* an even number of times — is a different statement, and is what feeds C1 and hence $\delta(G^D) \geq 1$. It does not give PAR.

### 5.4 UB42 — excluding $I = 42$

**Role.** C1 caps $I$ at 42, not at 40, so the value 42 must be excluded on its own. The manifest previously asserted twice that it is "excluded separately" without giving the argument. It is this.

**Status.** LEAN-KERNEL-CHECKED as `Exclude42.not_crossCount_42`, given the two named inputs X1 and X2; see §9. Also verified by exhaustive sweep, below.

$I = 42$ forces $|E^D| = 7$. With $\delta(G^D) \geq 1$ on $7 + 7$ vertices and only 7 edges, every vertex has degree exactly 1, so $G^D$ is a *perfect matching*. Its components are therefore the 7 matched pairs: each component contains exactly one side of $P$ and one of $Q$. In particular no two sides of $P$ share a component.

Next, both polygons are *exceptional*. Lemma 2 with $|S| = 4$ would force $|E^D| \geq 7 + (7 - 4) = 10 > 7$, so neither $P$ nor $Q$ admits an uncrossable 4-set. Hence the crossing-set data of §5.2 applies to each.

Now fix $j$. Since $G^D$ is a perfect matching, $N(q_j) = \{i_j\}$ is a singleton, so the set of sides of $P$ that $q_j$ *crosses* is $\{0, \ldots, 6\} \setminus \{i_j\}$, of size 6. Compatibility requires this crossing set to be contained in a maximal crossing set of $P$'s class. Every exceptional class has exactly five maximal crossing sets, each of size 6 and of the form $\{0, \ldots, 6\} \setminus \{m\}$ for $m \in M_P$, with $|M_P| = 5$. Two 6-subsets of a 7-set nest only if they are equal, so $i_j \in M_P$. This holds for every $j$, and the $i_j$ are distinct because $G^D$ is a matching, so all seven sides of $P$ lie in $M_P$ — contradicting $|M_P| = 5$. Hence $I \neq 42$.

*Remark.* Verified by exhaustive sweep over all $7! = 5{,}040$ perfect matchings against all 14 profile classes: 0 survive. Re-run against the independent profile reconstruction of §7 (whose class labels differ from the shipped ones by a dihedral relabelling of side indices, matched 14/14): also 0. The two invariants consumed — $|M| = 5$, and maximal crossing sets $= \{\text{FULL} \setminus \{m\} : m \in M\}$ — were recomputed from the class data rather than read from the stored invariant block, and hold for all 14 classes in both reconstructions.

*Remark* (This branch does not depend on A7-EXT). Earlier versions of this manifest excluded $I = 42$ through A7-EXT: a perfect matching makes every consecutive pair relevant and every index-non-adjacent pair span four distinct components, so $H$ is the complement of the 7-cycle, which contains the triangle $e_0e_2e_4$ and is not bipartite. That argument is correct but unnecessary. The route above consumes only C1, LA and the exceptional-class invariants EX, all of which are elementary and order-type based, so no further external lemma is imported — in particular not [5, Lemma 3], whose four-page case analysis this project has not audited and which an earlier draft wrongly treated as the only available reroute.

The practical consequence is that A7-EXT is load-bearing at exactly *one* point in the proof: the $I = 40$ endgame. It cannot be dropped there — with the filter off the pipeline ends at 2,187 surviving triples on 412 graphs, not 0 (§6).

*Remark.* The shipped $|E^D| = 7$ sweep reproduces the older route independently: $5{,}040 \to 14$ (A5-C) $\to 0$ (A7-EXT). The companion $|E^D| = 8$ sweep gives $476{,}280 \to 1{,}078 \to 0$, for $I = 41$, which parity already excludes. Both were a by-product of the historical enumeration bug of Remark 6.1 and are corroborative rather than load-bearing.

## 6 Computational reduction

### 6.1 The universe of candidate graphs

There are $7 \times 7 = 49$ side pairs, and $I + |E^D| = 49$, so $I = 40$ forces $|E^D| = 9$. By C1 the minimum degree is at least 1 on both sides. The number of bipartite graphs on $7 + 7$ vertices with exactly 9 edges and minimum degree at least 1 is

$$\sum_{j=0}^{7} (-1)^j \binom{7}{j} [x^9]\big((1+x)^{7-j} - 1\big)^7 \;=\; 13{,}003{,}620,$$

by inclusion–exclusion over uncovered $Q$-sides. This closed form was evaluated in exact integer arithmetic independently of both enumerations.

*Remark* (A historical enumeration bug). An earlier implementation reported 13,484,940 as "the number of 9-edge graphs". Its recursion never forced the edge budget to be exhausted at the leaf, so it enumerated $|E^D| \in \{7, 8, 9\}$:

$$5{,}040 \;+\; 476{,}280 \;+\; 13{,}003{,}620 \;=\; 13{,}484{,}940.$$

The searched universe was therefore a *superset* of what was needed. No candidate was missed and the exclusion was not invalidated — only the reported figure was wrong, and the $|E^D| = 7$ and $|E^D| = 8$ cases were swept for free. The bug nonetheless motivated the independent reimplementation of §7.

## 6.2 The filter pipeline

| Count | Filter applied next | Justification | Checked by |
|---|---|---|---|
| 13,003,620 | AKR L5, consecutive pairs | A5-C | Python, C++ |
| 42,238 | AKR L7 $\Rightarrow$ $H$ bipartite | A7-EXT + A7-BRIDGE | Python, C++ |
| 24,990 | crossing-set compatibility | elementary (order type) | Python, C++ |
| 2,300 | Černý L2, class-dependent | C2 | Python, C++ |
| 121 | *(change of counted object)* | — | Python, C++ |
| 225 | AKR L5, avoiding pairs | A5-A | Python, certificate, Lean |
| 0 | — | — | — |

*Remark* (Which filters the second implementation covers). The **Checked by** column above says "Python, C++" for the first four filters and *omits* C++ for the final avoiding-pair stage. That is not an oversight: Implementation B **deliberately does not implement the avoiding-pair akr L5 filter** (`reduction.cpp`:20; `SPEC.md` §6; `"final_avoiding_pair_filter_applied": false` in its output), nor Černý L3. It reproduces the reduction down to the 225-triple set and stops.

So the step that actually refutes — 225 $\to$ 0 — has *certificate* and *Lean* independence but **no second implementation**. Those two are real: `check_final_certificate.py` shares no code with the generator, and `Endgame.lean` kernel-checks each of the 225 cases. But the phrase "two implementations agree" covers filters (a)–(d) only. Recorded as K-20.

*Remark* (The intermediate counts are artifacts; only the final 0 is a claim). The figures 2,300 and 121 depend on which representative labelling of each exceptional class the pipeline carries: a superseded pipeline reported 3,522 and 64 at those same two stages. Nothing is wrong with either — the stages count labelled objects, and the labelling is a choice — but the comparison of two runs at an intermediate stage is meaningless unless the labelling is fixed. What *is* labelling-invariant, and what the proof actually needs, is the terminal emptiness, which was re-derived per ordered class pair: 196 ordered pairs $\times$ 3 labellings, 0 survivors in every case. Read every intermediate count in this manifest accordingly.

*Remark* (The 121 $\to$ 225 step is not a filter). This is a change in what is being counted, and it should not be read as a reduction. The figures 13,003,620 through 121 count *graphs*; the figure 225 counts *triples* $(\mathrm{class}_P, \mathrm{class}_Q, G)$. The 225 triples live on exactly those same 121 graphs: a graph may be compatible with several exceptional classes on either side, and $225 = \sum_G |ok_P(G)| \cdot |ok_Q(G)|$ over the 121 surviving graphs. Every one of the 121 graphs has degree sequence $(2, 2, 1, 1, 1, 1)$ on both sides, and only 9 of the 14 classes occur.

*Remark* (A7-EXT cannot be dropped from this branch). The filter at 42,238 is the only place A7-EXT enters the proof (Remark 5.4), which raises the question of whether it is needed at all. It is. Re-running the whole pipeline with that filter switched off gives

$$13{,}003{,}620 \to 42{,}238 \to 6{,}376 \to 533 \to 2{,}412 \to \mathbf{2{,}187}$$

surviving triples, on 412 graphs — not 0. The structure is sharp: the 533 graphs surviving Černý L2 without A7-EXT split as exactly 121 $H$-bipartite + 412 non-bipartite; the 121 are identical to the shipped certificate's graphs; and the avoiding-pair filter A5-A kills all 121 and *none* of the 412. On the residual the two filters are perfectly complementary, so no reordering helps. No already-trusted ingredient substitutes: compatibility is already applied at full strength, and Černý Lemma 3 kills 0 of the 412.

This is evidence about the pipeline, not about AKR Lemma 7: it shows the present toolkit cannot avoid A7-EXT, not that A7-EXT is mathematically indispensable.

*Remark* (Černý Lemma 3 is not a dependency). An earlier pipeline also applied Černý Lemma 3 between A7-EXT and the compatibility filter, reducing 24,990 to 9,702. Running the pipeline with and without it produces *identical* results at every subsequent stage and an identical set of 225 terminal triples: the graphs Lemma 3 removes are all removed by crossing-set compatibility in any case. Lemma 3 is therefore *not* a logical dependency of this proof and is omitted throughout.

# 7 Independent implementations

**Python (reference).** The readable implementation: recursion over $P$-side neighbourhoods with a running edge budget, union-find components, DFS 2-colouring for bipartiteness, a 128-entry lookup table for compatibility, and iteration over all $\binom{7}{4}$ subsets for Černý L2.

**C++17 (independent reduction).** Written from a written specification, with every major algorithm chosen differently: enumeration transposed to $Q$-side columns and partitioned by an explicit degree sequence rather than an edge budget; graphs keyed as a packed 49-bit word; components by iterative bitmask closure; bipartiteness by union-find *with parity*; compatibility by direct containment in the five maximal crossing sets; Černý L2 by enumeration of systems of distinct representatives.

**Agreement.** Both produce the same universe 13,003,620; the same intermediate counts 42,238, 24,990, 2,300, 121; the same 121 graphs; and the same canonical set of 225 terminal triples, with *zero set difference* in either direction.

*Remark* (Scope of the independence claim). This is **implementation-independent in algorithm and data representation, but not fully author- or specification-independent**: the second implementation was written by an agent that had already seen the first implementation and its specification. The agreement is therefore strong evidence against transcription and coding errors, and against a missed branch in either enumeration, but it is *not* evidence against a shared misreading of the published lemmas.

# 8 Certificates

A *certificate* here is a file that lets a referee re-derive the terminal contradictions without trusting — or rerunning — the exhaustive search. For each of the 225 terminal triples it stores the class identifiers, a canonical graph encoding (seven neighbourhood bitmasks), the degree sequences, the component partitions, the relevant side pair and its kind, both neighbourhoods, the two set differences, and the list of cyclic adjacencies found, which is empty in every case.

The standalone checker recomputes every quantity from the canonical encoding: neighbourhoods, degree sequences, components (by breadth-first search, where the generator used union-find), the set differences, and the adjacency test. The certificate's convenience fields are treated as claims to be checked, never as input.

*Remark* (Component labels are compared modulo relabelling). Component identifiers are arbitrary. The checker compares the induced *partitions*, not the integer labels. An earlier version of the comparison compared raw labels and reported spurious failures on 217 of 225 cases; the check was corrected to be relabelling-invariant.

**Adversarial controls.** The checker was validated against ten deliberate corruptions, all detected: a case marked as surviving, a pair not avoiding for its class, falsified neighbourhoods, falsified difference sets, a falsified adjacency list, a graph edited to break the edge count or minimum degree, a wrong degree sequence, a wrong component partition, and — the essential control — an *honestly constructed* witness using a different avoiding pair of the same class for which Lemma 5 genuinely *is* satisfied. That last case is rejected with a message naming the satisfying adjacency, which confirms the checker does not simply reject everything.

## A worked terminal case

Take the first certificate entry. The graph is given by the $P$-side neighbourhoods

$$N(p_0), \ldots, N(p_6) \;=\; \{0\}, \{0\}, \{1\}, \{2,4\}, \{3\}, \{4\}, \{5,6\},$$

with $P$ assigned exceptional class 12, whose $M$-set is $\{0,2,3,4,6\}$ and whose avoiding pairs are $(0,2)$ and $(4,6)$. The sides $p_4$ and $p_6$ are avoiding and lie in different components. Now

$$N(p_4) = \{3\}, \qquad N(p_6) = \{5,6\},$$

so $N(p_4) \setminus N(p_6) = \{3\}$ and $N(p_6) \setminus N(p_4) = \{5,6\}$. In $\mathbb{Z}_7$ the element 3 is cyclically adjacent only to 2 and 4, and neither lies in $\{5,6\}$. So (1) fails, contradicting A5-A. All 225 cases have this shape.

For contrast, the *other* avoiding pair $(0,2)$ of the same class on the same graph gives $N(p_0) \setminus N(p_2) = \{0\}$ and $N(p_2) \setminus N(p_0) = \{1\}$, and 0 *is* adjacent to 1 — so (1) holds there. The condition is genuinely discriminating: across the 225 certificate graphs, 2,016 of the 4,725 side pairs satisfy it.

# 9 Lean trust boundary

Environment: Lean 4.33.1; Mathlib tag `v4.33.1`, revision `0df444a3` [8]. Build: `lake build`.

**Twenty-five** libraries are declared in `lakefile.toml` and all twenty-five build: 1,395 jobs, exit 0, verified 2026-09-14. None contains `sorry`, `admit`, a project-specific `axiom`, `unsafe`, `partial`, or `native_decide`; this was verified by scanning the sources with comments stripped. All `decide` calls are ordinary kernel reduction, so nothing rests on the compiler.

*Remark* (Read §1.1 first). Twenty-five libraries is enough to create a false impression of coverage. The target statement is *not* proved; it is *stated*, with two hypotheses. The value of the development is that it makes the residue precise and machine-readable, not that it closes it.

## The twenty-five libraries

Grouped by layer. "core" in the **Base** column means the library and everything it imports are Mathlib-free, so Mathlib is outside its trust base; thirteen of the twenty-five are, and five of those import literally nothing (`LowerBound`, `GeometryDefs`, `EndgameDefs`, `Lemma7Bridge`, `CernyL2Interface`). The other twelve reach Mathlib, and exactly one — `PointSetSemantics` — reaches the reals.

| Library | Base | ID | What it proves |
|---|---|---|---|
| *Certificate and elementary geometry* | | | |
| `LowerBound` | core | LEAN-LB | the explicit pair: 38 crossings over provably all 49 side pairs; orientation-level simplicity and general position |
| `GeometryDefs` | core | LEAN-GEO | orientation predicates, lines, cyclic vertex access |
| `SegmentGeometry` | core | LEAN-GEO | a closed polygon meets a line evenly; hence $\delta(G^D) \geq 1$ and Lemma A's $h_2$, *derived* rather than imported |
| `LemmaA` | Mathlib | LEAN-LA | the counting core of Lemma A and the parity upgrade, as an implication |
| `LemmaABridge` | Mathlib | X3 | Lemma A's `Finset` count equals the mask-level `degSum`; hence "$\lvert E^D\rvert < 10 \Rightarrow P$ exceptional" |
| *Geometry-to-combinatorics bridge* | | | |
| `DisjointnessGraph` | core | I1, I2a | `Config`; `toMasks` and its orientation-level spec; minimum degree on both sides |
| `PointSetSemantics` | Mathlib, $\mathbb{R}$ | I2b | `Disjoint0` iff the two closed sides are disjoint *as point sets in* $\mathbb{R}^2$ |
| *The stored endgame* | | | |
| `EndgameDefs` | core | — | the endgame predicates; `akr5Conclusion_iff_adjacent` |
| `EndgameData` | core | — | the 225 certificate records, generated from the committed JSON |
| `Endgame` | core | LEAN-END | all 225 cases fail the AKR-L5 condition; components recomputed in-kernel |
| `AvoidingFlag` | core | I3 (flag half) | the stored `avoiding` flag *equals* the class table's avoiding-pair predicate |
| *Filter transcriptions and their obligations* | | | |
| `Lemma7Bridge` | core | A7-BRIDGE | non-empty + pairwise-disjoint status sets $\Rightarrow H$ bipartite |
| `CernyL2Interface` | core | C2-BRIDGE | the L2 filter fires iff an *exact induced* perfect matching exists |
| `FilterSoundness` | core | E1–E8 | each filter's soundness obligation, as an implication from named external inputs |
| `ComponentLabelling` | core | E6 | a fuel-bounded closure labelling, equal labels iff connected, both directions |
| `BipartiteDecision` | core | E8a | a total fuel-bounded 2-colouring test, `true` iff bipartite, both directions |
| `CernyL2Derivation` | Mathlib | E5 | **attempted and blocked**; see §9 |
| *The enumeration obligations* | | | |
| `EnumerationDomain` | Mathlib | obl. (1) | a configuration with $I = 40$ lands in the search domain |
| `DomainCount` | Mathlib | obl. (2) | \|domain\| = 13,003,620 |
| `EnumerationComplete` | Mathlib | obl. (3) | the transcribed search recursion visits exactly the domain |
| `CounterexampleCount` | Mathlib | (1)+(2) | a counterexample is one of exactly 13,003,620 mask functions |
| *Upper-bound assembly* | | | |
| `Exclude42` | Mathlib | UB42 | $I \neq 42$, from X1 and X2 |

| Library | Base | ID | What it proves |
|---|---|---|---|
| `UpperBoundAssembly` | Mathlib | — | $I \leq 42$ unconditionally; $I \leq 38$ from parity, $I \neq 42$, $I \neq 40$ |
| `MainTheorem` | Mathlib | LEAN-MAIN | 38 is the greatest attained crossing count, given the hypotheses |
| `IntersectionParity` | Mathlib | PAR | $I$ is even for an arbitrary configuration; the final assembly |

## `LowerBound.lean` — LEAN-LB

**Imports: none.** Core Lean only; verified to compile with an empty `LEAN_PATH`, so Mathlib is not in its trust base. **Axioms:** `propext` only.

Twenty theorems, including `P_simple`, `Q_simple` (distinct vertices; no three of the seven collinear; consecutive sides non-collinear; no proper crossing of non-adjacent sides), `general_position` (no vertex of one polygon on a side-line of the other), `crossing_count_eq_38`, `per_P_side_counts` $= [6, 6, 6, 6, 6, 6, 2]$, `disjoint_count_eq_11` and `crossCount_add_disjointCount`. Crucially `allPairs_length` $= 49$, `allPairs_nodup` and `allPairs_complete` establish that the count of 38 is taken over provably all 49 side pairs.

**Not formalized:** the step from these orientation-level facts to "the closed polygonal curve is a simple closed curve in $\mathbb{R}^2$".

## `LemmaA.lean` — LEAN-LA

**Imports:** Mathlib (`Finset.Card`, `Fintype.Prod`, `Fintype.Card`, `Algebra.Ring.Parity`) and `LowerBound`. **Axioms:** `propext`, `Classical.choice`, `Quot.sound`.

Theorems `card_E_ge`, `card_E_ge_ten`, `ge_eleven_of_ge_ten_of_odd`, `card_E_ge_eleven`, `crossings_le_38`, together with a non-vacuity block (`Rcert_h1`, `Rcert_h2`, `Rcert_card_E` $= 11$, `nonvacuous`, `cert_I_le_38`, `hypotheses_are_restrictive`) showing the hypotheses are simultaneously satisfiable, the bound tight, and the empty relation excluded.

*Remark* (A defect in the earlier skeleton). An earlier, never-compiled version of this file did not have `Odd` in scope. Lean's auto-bound implicits silently turned it into a free variable, so `card_E_ge_eleven` and `crossings_le_38` elaborated as vacuous statements *with no error reported*. The file now sets `autoImplicit false`, which makes this class of failure impossible.

## `Endgame.lean` — LEAN-END

**Imports:** `EndgameData` only, which imports `EndgameDefs`, which imports nothing — core Lean throughout, confirmed standalone with bare `lean` and `LEAN_PATH=`. **Axioms:** `propext` and `Quot.sound` only; notably *no* `Classical.choice`.

Principal theorems:

- `card_certificates` : the list has length 225;
- `keys_distinct` : no duplicate records under the canonical key;
- `all_225_cases_fail_AKR5_requirement` and `all_225_cases_fail_AKR5_conclusion`;
- `all_225_pairs_in_different_components` : component separation is *recomputed in the kernel* via a separator argument, not taken from the stored data;
- `akr5Conclusion_iff_adjacent` (proved one level down, in `EndgameDefs.lean`, in the same namespace) : the published conclusion and the implemented predicate are equivalent; this depends on no axioms at all;
- `akr5_not_vacuous` : 2,016 of the 4,725 side pairs in these graphs *do* satisfy the condition.

The Lean data file is generated deterministically from the committed certificate JSON, whose hash is embedded in the generated header.

*Remark* (The `avoiding` flag is no longer trusted — I3). `Case.avoiding` is a Boolean copied out of the certificate JSON field `witness.pair_kind`. Until 2026-09-14 Lean simply *trusted* it, and `Endgame.lean`'s own caution notes recorded that a record carrying a wrong flag would satisfy every theorem in the file.

`AvoidingFlag.lean` closes that. `all_225_avoiding_flags_eq` proves, by `decide` over all 225 records, that the stored flag *equals* $(\texttt{pairI}, \texttt{pairJ}) \in \texttt{avoidingPairs}(\texttt{classUsed})$ — an *equality*, so the `false` case is checked too, not the weaker implication that would leave a spuriously-`false` flag unexamined. The 14-class table is *generated from* `exceptional_profiles_14.json`, and was verified to match it class by class. Axioms: `propext`, `Quot.sound`.

What this removes is the flag as an *independent* failure mode. What it does *not* do is ground the flag in geometry: the correctness and completeness of the exceptional classification (EX), the correspondence between a record's class indices and an actual polygon, and a geometric definition of "avoiding" — of which this development has none — all remain open.

Two supporting facts, recomputed from the committed Lean data rather than from the pipeline that produced it: `classUsed` = `classP` in all 225 records, so the avoiding pair is always drawn from the $P$-side class; and only 9 of the 14 classes ever occur, covering exactly $81 = 9 \times 9$ distinct class pairs.

## `Lemma7Bridge.lean` — A7-BRIDGE

**Imports: none.** Core Lean; builds standalone with an empty `LEAN_PATH`. **Axioms: none at all** — every theorem reports *does not depend on any axioms*, not even `propext`. One documented exception, added 2026-09-14: the classical corollary `disjoint_statuses_imply_bipartite_classical`, which removes the decidability hypothesis, depends on `Classical.choice`. That hypothesis is a formalisation convenience, *not* a mathematical or geometric assumption.

Formalises only the implication of §4.5. The two hypotheses are `hne` : $\forall e, \exists s,\ S\,e\,s$ and `hdisj` : $\forall e\,e'\,s,\ \mathrm{Adj}\,e\,e' \to S\,e\,s \to S\,e'\,s \to \bot$. Theorems: `adj_irrefl` (irreflexivity is derived, not assumed), `isolated_of_both`, `status_unique_of_adj`, `statusIs_of_adj`, `opposite_of_adj`, `disjoint_statuses_imply_bipartite` (main), and a partition-form variant carrying no decidability hypothesis. Non-vacuity is proved rather than asserted: a concrete model with a real edge; a both-status vertex shown isolated; and a triangle shown both non-bipartite and admitting no valid status assignment, so the hypotheses genuinely constrain the graph.

**AKR Lemma 7 does not appear** — not as an axiom, not by name. Grepping the code with comments stripped finds no occurrence of AKR or Observation.

## `CernyL2Interface.lean` — C2-BRIDGE

**Imports: none. Axioms:** the three standard ones; the pure `decide` facts are `propext`-only.

Proves `l2FiresB_iff`: the *Lean transcription* of the filter fires iff an *exact induced* perfect matching exists, the single clause $\mathrm{Edge}\,(s_t)\,(q_u) \leftrightarrow t = u$ carrying both halves — all four matching edges present, all twelve cross pairs absent. The decisive fixtures: a matching plus an extra edge *inside* the 4-block is no longer the forbidden pattern *for that block*, while a matching plus an edge *outside* the block **still** fires — so the predicate is certified to be about the induced subgraph rather than about degrees. Note the first is a per-fixture fact: an extra edge does not in general stop the filter firing, since *another* block can still be an exact induced matching.

*Remark* (Limits of C2-BRIDGE). It certifies the Lean transcription of the filter, *not* the Python/C++ source: there is no extraction, and the correspondence remains a human-eyeball link. One representational shortcut is unverified — the code computes $\mathrm{popcount}(N_s \wedge S)$ where the Lean definition sums four indicator bits; these agree exactly when the four indices are distinct, which the enumeration enforces, but the bit-level identity is not proved. And it says nothing about Černý Lemma 2 itself, nor about the per-class Lemma-2 window sets, which remain fully external.

## `GeometryDefs.lean` + `SegmentGeometry.lean` — LEAN-GEO

**Imports: none** (core Lean). **Axioms:** `propext`, `Quot.sound` — *no* `Classical.choice` across all 62 theorems.

The first genuinely geometric milestone: Černý L1's required consequence and Lemma A's hypothesis `h2` are *derived* from a parity argument rather than imported. The chain is `cycChanges_even` → `metCount_even` → `metCount_lt_of_odd` → `min_disjointness_degree_fin7` → `lemmaA_h2_fin7`: a closed polygonal curve meets a line an even number of times, so a side of $P$ cannot cross all seven sides of $Q$. The connection lemmas to `LowerBound`'s definitions hold by `rfl`, so the two developments are not merely analogous but definitionally identical.

## `DisjointnessGraph.lean` — I2a

**Imports:** `GeometryDefs`, `LowerBound`, `EndgameDefs`, `SegmentGeometry` (all Mathlib-free). **Axioms:** `propext`, `Quot.sound`.

The geometry-to-endgame bridge. `Config` packages two simple heptagons in general position; `toMasks_spec` proves that bit $j$ of mask $i$ holds *iff* side $i$ of $P$ does not properly cross side $j$ of $Q$, and `min_degree_both_sides` supplies minimum degree $\geq 1$ on both sides by *application* of the general theorem, not by `decide` on the certificate.

*Remark* (Why the interface is split into I2a and I2b). The interface as originally written stated `toMasks_spec` against *point-set* disjointness of two segments in $\mathbb{R}^2$. That is not statable in a Mathlib-free development: `Pt` is $\mathbb{Z} \times \mathbb{Z}$ and `side` $P\,i$ is never defined as a set. Stating it needs Mathlib's `segment` and the reals. The interface was therefore split: I2a, the orientation-level form proved here and consumed by everything downstream, and I2b, the single point-set lemma `DisjointO` $K\,i\,j \iff \mathrm{Disjoint}(\mathrm{seg}_P\,i, \mathrm{seg}_Q\,j)$, which is quarantined in one leaf library so that the rest of the development keeps its Mathlib-free base. `Endgame.lean` consumes only the *function* `toMasks`, never the specification, so no endgame theorem depends on either half.

## `PointSetSemantics.lean` — I2b

**Imports:** `DisjointnessGraph` and Mathlib (`Analysis.Convex.Segment`, `Data.Real.Basic`). **Axioms:** the three standard ones.

LEAN-KERNEL-CHECKED as of 2026-09-14, and this closes I2b. The integer plane is embedded coordinatewise in $\mathbb{R} \times \mathbb{R}$ and `disjoint0_iff_disjoint` proves, *in both directions and with no hypotheses beyond `Config`'s own general-position fields*,

$$\texttt{Disjoint0}\ K\, i\, j \iff \mathrm{Disjoint}\big(\texttt{segment}_{\mathbb{R}}\, p_i,\ \texttt{segment}_{\mathbb{R}}\, q_j\big).$$

Composed with I2a this gives `toMasks_pointSet_spec`: the stored mask bit means what it is supposed to mean about point sets in the real plane. Non-vacuity is proved on the certificate pair in both directions — one disjoint pair, and one properly crossing pair together with an explicit common real point.

**What this does *not* do.** It makes *side* disjointness faithful to $\mathbb{R}^2$. It says nothing about the polygon as a curve: the Jordan-curve step — that the closed polygonal chain of `Simple7` bounds a region, which is what "simple polygon" means in $f(7,7)$ — is still not formalised anywhere, and neither is the general-position reduction GP. Those are the two steps of §1.1, and this library reduces neither.

## The enumeration obligations — `EnumerationDomain`, `DomainCount`, `EnumerationComplete`, `CounterexampleCount`

**Imports:** `DisjointnessGraph`, `Endgame`, and Mathlib (`BigOperators`, `Polynomial.Coeff`, `InclusionExclusion`, `Nat.Choose.Sum`, `Finset.Powerset`, `Fintype.Pi`). **Axioms:** `propext`, `Quot.sound` throughout `EnumerationDomain`'s Part A — *no* `Classical.choice`, and no Mathlib result is used in substance there; the counting parts add `Classical.choice` through Mathlib's `Finset` and `Polynomial` machinery.

"Completeness of the enumeration" is four claims, not one:

1. a hypothetical counterexample produces a graph satisfying the search conditions;
2. the abstract search domain has 13,003,620 elements;
3. the search *actually visits* every element of that domain;
4. every valid configuration survives each filter until it reaches a terminal case proved impossible.

**(1) is proved, in full.** `toMasks_inDomain_of_crossCount_eq_40` states that for an *arbitrary* `Config` — not the certificate pair — $\texttt{crossCountC}\, K = 40$ implies $\texttt{InDomain}\,(\texttt{toMasks}\, K)$: seven 7-bit masks, nine edges, minimum degree $\geq 1$ on both sides. The route is `crossCountC_add_degSum` (that is, $I + |E^D| = 49$ proved for *every* configuration rather than checked on one) together with I2a's minimum-degree result. The hypothesis $I = 40$ is an *input*; why a counterexample must have exactly 40 crossings is Lemma A plus parity, argued elsewhere. Definitionally, `degSum_eq_edgeCount` holds by `rfl`, so `InDomain`'s edge clause *is* the quantity the certificate checker recomputes.

**(2) is proved.** `DomainCount.domainFinset_card` closes it:

$$|\mathrm{domainFinset}| = 13{,}003{,}620,$$

supplying the two steps `EnumerationDomain.lean` had left open — that the $t$-th inclusion–exclusion term depends only on $|t|$ (the source of the $\binom{7}{j}$), and that for $|t| = j$ it equals $[x^9]\big((1+x)^{7-j} - 1\big)^7$. `CounterexampleCount` then joins (1) and (2) into the single sentence *a hypothetical counterexample is one of exactly* 13,003,620 *mask functions*. The count is proved, not evaluated: nothing in the kernel enumerates the set.

**(3) is proved, about the transcription.** `EnumerationComplete.enumerate_complete` proves, by induction on the recursion depth, that the Lean transcription of `python/elimination/audit_enum.py`'s `rec` emits every element of `InDomain` — no bound, no special case — and `enumerate_sound` proves the converse, so the recursion visits *exactly* the domain. The pruning bound $\min(3, \texttt{budget} - \texttt{rem})$ is transcribed verbatim and then *proved harmless* rather than dropped. The one presentational change is that the loop counter counts down, which makes the recursion structural: no `partial`, no well-founded recursion, no `decide` on the output list.

**(4) is stated, not proved; and none of this is about the shipped programs.** Obligation (4) is `FilterSoundness.lean`, whose eight external inputs are tabulated in §9; six of the eight are open. Independently of that, obligations (3) and (4) alike are statements about *Lean transcriptions* written by hand from the Python and C++ sources. Nothing in Lean reads, compiles, or cross-checks either program; there is no extraction. So "the search recursion is complete" means "the Lean function is complete, and it was transcribed by eye from the Python".

*Remark* (A degree-sequence question, raised and closed). `Endgame.graphShapeOK` demands degree sequence $(2,2,1,1,1,1,1)$ on both sides, whereas seven positive integers summing to nine also admit $\{3,1,1,1,1,1,1\}$; that branch is genuinely non-empty in the domain (`sampleM`). This looks like an unchecked filter and is not one: `graphShapeOK` is a *descriptive* check on the 225 survivors, not a search precondition, and both implementations enumerate the degree-3 branch, where it dies in the L5/L7/L3 filters. Recorded as K-17.

## `FilterSoundness.lean` and its two discharges — E1–E8

**Imports:** `DisjointnessGraph`, `Lemma7Bridge`, `CernyL2Interface` — all Mathlib-free, so this library and the two that discharge parts of it stay outside Mathlib's trust base. **Axioms:** the three standard ones; many declarations depend on none.

This is where the pipeline's honesty is enforced. For each of the five filters it states the obligation *a valid configuration's mask function is not discarded by this filter*, and proves that obligation as an implication from explicitly named external inputs. It proves none of those inputs, and — the point of the design — it does not declare them as `axiom`s either, so `#print axioms` stays clean and a reader can see at each theorem exactly which import is carrying the weight.

| ID | External input | Status | Where |
|---|---|---|---|
| E1 | AKR L5 for consecutive pairs | OPEN | hypothesis `hAKR5` of `filterA_sound` |
| E2 | AKR L5 for avoiding pairs, *and* that the stored list $AV$ really consists of avoiding pairs | OPEN | `hAVlist`, `hAKR5` of `filterE_sound` |
| E3 | AKR L7 in the project's form: non-empty status sets, disjoint across $H$-edges | OPEN (= A7-EXT) | `hne`, `hdisj` of `filterB_sound` |
| E4 | the order-type fact behind filter (c): $q_j$'s crossing set sits in a maximal crossing set of $P$'s class | OPEN | `hcross` of `filterC_sound`; reused as X2 |
| E5 | Černý L2: no Lemma-2 window carries an exact induced perfect matching | OPEN; attempted, **blocked** | `hC2` of `filterD_sound`; §9 |
| E6 | correctness of the component labelling | DISCHARGED | `ComponentLabelling.lean` |
| E7 | $K$'s heptagons are among the 14 enumerated classes, and the stored class data is correct for $K$ | OPEN — **the classification itself** | `hmem` of `filterCDE_sound` |
| E8a | bipartiteness is decided by a correct algorithm | DISCHARGED | `BipartiteDecision.lean` |
| E8b | *the shipped programs' routine* is such an algorithm | OPEN (transcription) | no extraction exists |

So **six** of the eight — E1–E5 and E7 — are open outright, and the transcription half E8b is open as well. Of the six, E3 *is* A7-EXT, the principal item of §4.5, and E7 is the exceptional classification, the largest single unformalized computational object in the project.

**E6, discharged.** `ComponentLabelling.lean` defines a concrete labelling of the 14-vertex disjointness graph by a fuel-bounded closure iteration — no `partial`, no well-founded recursion — and proves *equal labels iff connected*, in both directions. The asymmetry the file makes explicit is worth repeating: filters (a) and (e) need only "different labels ⇒ really different components", because a labelling that merges components makes them test fewer pairs; filter (b) needs the *iff*, because $H$'s edges are defined by label distinctness, so a labelling that splits a component gives $H$ more edges, and bipartiteness of the true $H$ says nothing about a supergraph of it.

**E8, half discharged.** `BipartiteDecision.lean` defines a total, fuel-bounded 2-colouring test and proves `bipTest = true` iff the graph admits a proper 2-colouring, then instantiates it at $H$. The "standard algorithmic fact" E8 named is now a kernel-checked theorem. What remains is E8b: the shipped C++ uses a parity disjoint-set union and the Python a DFS, whereas the verified Lean procedure is a fixed-point colour propagation. It is in the same family, not a byte-for-byte transcription, and no Lean code reads either program.

*Remark* (Transcription, not extraction — the gap common to every filter). `l5TestB`, `HAdj`, `CompatOk`, `L2Ok` and `L5PairsOk` were written *by hand* from `python/elimination/audit_enum.py` and `independent/implementation_b/reduction.cpp`. Where a theorem in this development says "the filter", read "the Lean transcription of the filter". This applies to C2-BRIDGE, to E8, to obligation (3), and to every filter-soundness statement; it is a single systematic gap, not several independent ones, and closing it would require extraction or a verified compiler, neither of which is present.

*Remark* (Two divergences between the two implementations, recorded here). `audit_enum.py` runs a *sixth* filter, Černý Lemma 3, between (b) and (c); its soundness obligation is a genuine sixth obligation and is not stated in `FilterSoundness.lean`. It is also not a dependency: the pipeline gives an identical terminal set with and without it. And `reduction.cpp` does not implement filter (e) at all (Remark 6.2). So the intersection of what the two programs check is filters (a)–(d).

## `Exclude42` and X1–X3

**Imports:** `EnumerationDomain`, `FilterSoundness`, Mathlib. **Axioms:** the three standard ones.

`Exclude42.lean` formalises §5.4. Kernel-checked from its hypotheses: seven edges plus minimum degree 1 forces every degree to be exactly 1 on either side (via the transpose double-count); a degree-1 mask has at most one set bit; the partner function $j \mapsto i_j$ is well defined and injective; and the nesting step, that a 6-set $\{i : i \neq i_j\}$ inside a 6-set $\{i : i \neq m\}$ forces $m = i_j$. The corollary `not_crossCount_42` is stated at the level of a geometric `Config`, and $I + |E^D| = 49$ and $\delta(G^D) \geq 1$ are *not* hypotheses of it — they are imported theorems.

Three external inputs were named. Two remain:

- X1, the exceptional-class invariant: $|M_P| = 5$ and the maximal crossing sets are exactly $\{\text{FULL} \setminus \{m\} : m \in M_P\}$. This is part of EX, and it is *not* proved in Lean.
- X2, the order-type fact behind filter (c) — identical to E4, reused verbatim rather than restated.
- X3, that $P$ is exceptional, DISCHARGED 2026-09-14 by `LemmaABridge.lean`: `card_E_eq_degSum` identifies Lemma A's `Finset` count with the mask-level `degSum`, after which `degSum` $< 10$ yields "no uncrossable 4-set" by application of Lemma A rather than by assumption.

## `IntersectionParity`, `UpperBoundAssembly`, `MainTheorem` — the assembly

**Axioms:** the three standard ones throughout.

`UpperBoundAssembly` proves $I \leq 42$ *unconditionally* for every `Config`, by applying $I + |E^D| = 49$ and the minimum-degree theorem — so the project's use of Černý L1 is, at this point, a consequence of its own parity argument rather than an import. It then proves that parity, $I \neq 42$ and $I \neq 40$ suffice for $I \leq 38$, and supplies a variant in which the $I = 42$ branch is discharged through `Exclude42` instead of assumed. `MainTheorem` writes down the target sentence; `IntersectionParity` discharges parity (PAR, §5.3) and assembles `isGreatest_38_final`, quoted in §1.1.

Non-vacuity is proved rather than asserted at each stage: the certificate pair satisfies parity and both exclusions, and the bound is attained there, so the assembly is not a statement about the empty set and 38 is not slack.

*Remark* (The one thing `hclass` is not known to be). `hclass` asserts the exceptional-class data for *every* `Config`, not only for the ones the reductio reaches. That is stronger than the mathematical claim it stands for, which is conditioned on $I = 40$ (or, in `Exclude42`, on $I = 42$). Nothing in the development proves `hclass` *satisfiable*, and no non-vacuity witness for it is supplied. A reader should therefore treat `isGreatest_38_final` as a statement of the residue, not as evidence that the residue is consistent.

## E5 — attempted and blocked

`CernyL2Derivation.lean` is an *attempt* at E5, Černý Lemma 2's consequence, and it does not succeed. Its own header says so: "this file does not discharge E5". It must not be quoted as a proof of E5 or of Černý Lemma 2.

What it proves is strictly weaker than E5: an integer affine identity for the triangle determinant; that a positively oriented interior point stays in any strict positive half-plane containing the triangle's vertices; that simplicity excludes proper crossings even of adjacent sides; and the conditional statement "E5 holds if `CollisionObstruction` does", for the full forward-or-reverse window predicate. `CollisionObstruction` is the unresolved step, stated explicitly rather than assumed silently: the four complementary crossing patterns at an inside-triangle window must force a crossing between two sides of $Q$. That is a *simultaneous* segment realizability argument, and testing each selected segment separately cannot reach it — the file includes `localFixture`, a concrete configuration in which all sixteen required crossing and non-crossing entries hold, the triangle condition holds, the five-vertex path is simple, and cross general position holds, yet the selected segments cross each other. It is deliberately not a `Config`.

## UB40 is out of reach of the kernel

`hne40` — $I \neq 40$ — is the one hypothesis of `isGreatest_38_final` that is not an external lemma but a *computation*. The natural hope is that the Lean kernel could simply perform it. **It cannot**, and as of 2026-09-14 this is a settled negative, not an open engineering question. `native_decide` is excluded throughout the project because it adds `Lean.ofReduceBool` to the trust base; what follows concerns ordinary kernel reduction. Five routes were measured.

1. **Monolithic kernel evaluation — dead on memory.** A full traversal is 226,668,772 recursive calls. A ladder of partial runs fits user $\approx 0.135 + 2.385 \times 10^{-4} \cdot$ calls seconds, i.e. ≈15 CPU-hours, but peak resident set runs at 20–25 KB per call: 4.5–5.8 TB against 8 GB installed. And that is for *counting only*, with no filter evaluated.
2. **Chunked evaluation — memory solved, wrong problem.** Splitting by the first four rows holds memory near 2 GB (1,917,400 calls at 2.23 GB), which retires the 4.5 TB figure as an artifact of building one enormous term. But it does not help: the counting recursion memoises 226,668,772 states down to 1,925 distinct ones,

a sharing factor of ≈117,750, and counting was *already* proved by `DomainCount`. Filter evaluation consumes the whole graph, so none of that sharing applies.

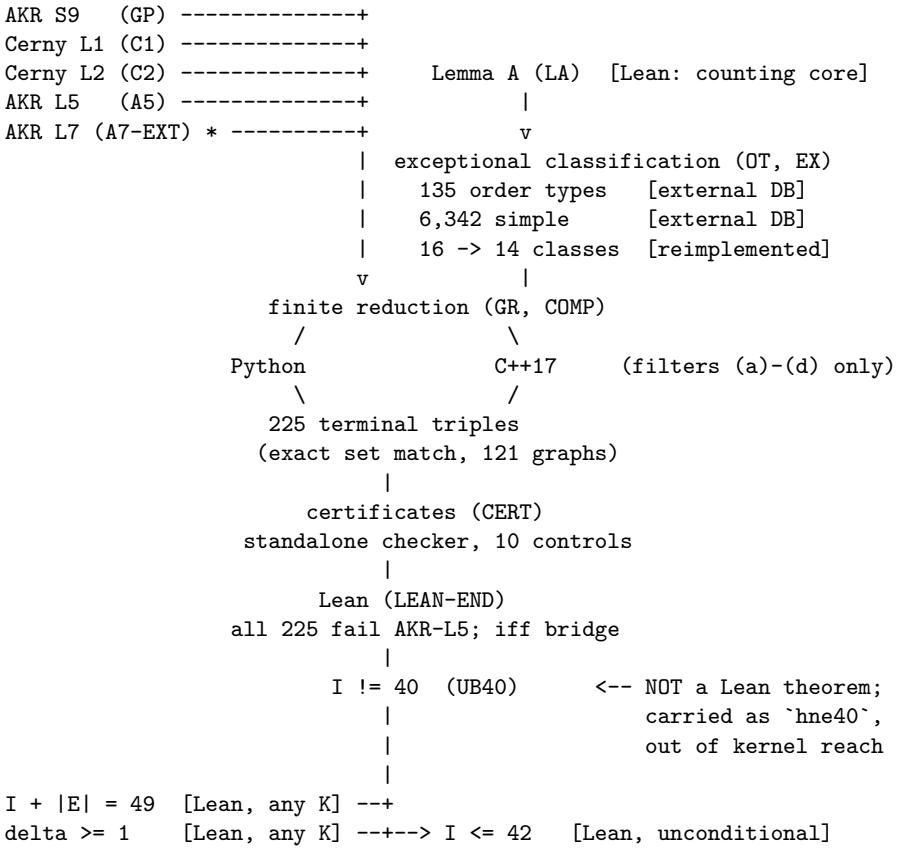
3. **Per-graph filter cost — the decisive measurement.** Forcing the kernel to evaluate the component labelling and bipartiteness test on 1, 10 and 40 graphs gives a marginal cost of ≈1.03 CPU-seconds per graph. Over 13,003,620 graphs that is ≈3,700 CPU-hours — about 154 days — for *two* of the five filters. The kernel-to-Python ratio is ≈70,000, not the ≈500 one would assume from arithmetic benchmarks. The obstacle is time, not memory.
4. **Re-encoding the filters for speed — 7.3×, need ≈75×.** Rewriting the labelling as unrolled `Nat` bitmask arithmetic with a 14-step fixed point (correctness checked against a reference first) takes 0.309 down to 0.042 CPU-seconds per graph on the same 40-graph benchmark. Projected over all five filters that is ≈1,000–1,500 CPU-hours; a feasible one-off needs about 50. The route was closed against a gate fixed in advance.
5. **Structural shortcuts — dead on arithmetic.** A UB42-style argument from cheap invariants cuts about 33%. Leading with the local, dynamic-programmable filters (crossing-set compatibility and Černý L2) still leaves 2,783,508 graphs of the 13 million for some class pair. Neither is within two orders of magnitude of enough.

**What is *not* settled by this.** These are measurements about evaluating the *present* 13,003,620-element domain in the kernel. A mathematical restructuring that shrinks the domain itself would change the picture entirely — but that is mathematics, not engineering, and nobody has one. The honest summary is that UB40 will not become a Lean theorem by computation on this architecture, and that if it is to be formalised at all it must first be made smaller.

> Key finite and combinatorial components are formally verified in Lean; the published geometric lemmas themselves remain external dependencies. In particular $f(7,7) = 38$ is **not** a Lean theorem — the strongest formalised statement carries the hypotheses `hclass` and `hne40` (§1.1) — the completeness of the 225-case list is not formalized, six of the eight external inputs E1–E8 are open, and every filter statement is about a hand-written Lean transcription rather than about the shipped programs.

# 10 Trust DAG

The upper bound splits into two independent branches at the top. Only the right branch touches A7-EXT, and only the right branch is unformalized.



```
                                |
I even (PAR)  [Lean, any K] --+
                                |
I != 42 (UB42) [Lean, given X1+X2] -- does NOT use A7-EXT
                                |
                                v
                           I <= 38   [Lean]
                                |
          explicit 38-example (LB38) [Lean: LEAN-LB]
                                |
                        f(7,7) = 38   (candidate)
                  -- modulo the Jordan-curve step and GP,
                     neither of which is formalized

* A7-EXT: VERY STRONGLY CORROBORATED, HUMAN REVIEW STILL DESIRABLE -- see
  the A7-EXT subsection.  The chain is
      A7-EXT  ==>  A7-BRIDGE  ==>  H bipartite
  where A7-EXT is the external reading of AKR Observation 6 / Lemma 7
  (adjacent H-vertices have disjoint status sets), and A7-BRIDGE is the
  project's own inference from that to bipartiteness -- the latter is
  LEAN-KERNEL-CHECKED, with the disjointness condition as an explicit
  hypothesis and no AKR axiom.
  If A7-EXT fails, the H-bipartite filter is unjustified and the
  reduction from 42,238 onwards must be redone; UB40 fails.
  A7-EXT enters at exactly one point, the I=40 endgame: the I != 42 branch
  is routed around it through crossing-set compatibility alone, and the
  filter cannot simply be dropped (without it 2,187 triples survive).
```

# 11 Known defects found during development

Listed because they calibrate how much checking this argument has had.

1. **Enumeration bug (13,484,940).** The leaf test omitted the edge-budget condition, so $|E^D| \in \{7, 8, 9\}$ was enumerated. *Effect on the theorem:* none — a superset was searched, so no candidate was missed. *Repair:* corrected leaf test; the buggy file is preserved unpatched as evidence. *Superseded by:* the corrected enumeration and the closed form of §6.
2. **Raw component-ID comparison.** A verification script compared component *labels* rather than the partitions they induce, and reported failures on 217 of 225 cases. *Effect:* none on the proof; it was a checker bug. *Repair:* compare partitions. *Superseded by:* the current relabelling-invariant checker.
3. **Lean `Odd` / `autoImplicit` defect.** In the earlier `LemmaA.lean` skeleton, `Odd` was out of scope and became a free implicit variable, so two theorems elaborated as vacuous statements with no error. *Effect:* none, because the file had never been compiled. *Repair:* rewritten with `autoImplicit false`. *Superseded by:* the current `LemmaA.lean`.
4. **Step 0 perturbation argument.** The project's own general-position proof is incorrect when $\mathrm{Cone}_P(v) \subset \mathrm{Cone}_Q(w)$, where its rule destroys the intersection point. *Effect on the theorem:* none, because the required statement follows from [2, §9.1]. *Repair:* removed from the active proof. *Superseded by:* GP, §4.1.
5. **Strict versus loose exceptional condition.** Early phrasing said "no line meets all four sides", which does not license what the computation tests. *Effect if it had been implemented:* severe — 2,268 exceptional objects instead of 16, and an infeasible search. *Repair:* the strict, vertex-avoiding form ($\star$). *Superseded by:* Lemma 2 and Remark 5.2.
6. **akr Observation 6 is false as printed** — a defect in a published paper, not in this project. Stated without proof in both versions and used throughout the case analysis of Lemma 7. The counterexample is a pair of genuine simple polygons,

$$P = (2,0), (3,4), (-2,0); \qquad Q = (4,-4), (-3,1), (0,-1), (3,3), (2,1),$$

verified in exact integer arithmetic and independently re-verified. Both are simple; the eight vertices are distinct; no three of the eight are collinear; no vertex of one lies on the other; no two of the eight sides are parallel; $I(P,Q) = 8$ is even — every general-position condition of [2, §3] holds. $\mathrm{CC}(p_0) \neq \mathrm{CC}(p_1)$ and $(p_0, q_0), (p_1, q_4) \in E^D$, so $q_0, q_4$ *is* associated to $p_0, p_1$ under the printed definition, and both cross non-edges hold as well; yet the cone dichotomy fails in both directions and the axis through the two apexes leaves $a, b, A, B$ in one closed half-plane. *Effect on this proof:* none. A pair failing the dichotomy is neither hooking

nor hooked, so it contributes nothing to $S(e)$ (Remark 4.5). *Effect on the manifest:* its earlier gloss of the narrow notion as "edges + cross non-edges" was wrong, since those non-edges are automatic and the dichotomy is what carries the content; corrected in §4.5.

7. **The A7 encoder's coverage argument was wrong.** The oriented-matroid case exhaustion reduced the four $Q$-adjacency cases to one representative by a symmetry orbit. That is valid for a $Q$-adjacency in isolation and *invalid* in combination with a $P$-adjacency: the relabelling generators act in tandem, so the combined types split into *four* orbits and only one was solved. Worse, that one was vacuous — its base pattern is itself UNSAT, so every PQ-adj verdict, including the two that matter, held for no reason. *Effect on the theorem:* none established either way at the time; the conclusion survived, the evidence for it did not. *Repair:* all 25 combined types enumerated with no symmetry reduction and solved separately, 125 certificates, hash-bound to their formulas. *Aggravating circumstance, recorded because it is the point of this section:* an earlier hostile review had checked the orbit structure for a $Q$-adjacency alone and endorsed the argument as sound. *Superseded by:* §4.5.
8. **Intermediate pipeline counts are labelling-dependent artifacts.** The figures 2,300 and 121 depend on which representative labelling of each exceptional class the pipeline happens to carry; a superseded pipeline reported 3,522 and 64 for the same two stages. *Effect on the theorem:* none — the terminal emptiness is labelling-invariant and was re-derived per ordered class pair (196 pairs, 3 labellings, 0 survivors in every case). *Standing consequence:* every intermediate count quoted in this manifest should be read as an artifact of a particular run, and only the final 0 as a claim.

# 12 Remaining work

In priority order. The first three are the binding constraint; the rest is optional polish.

1. **External human review of A7-EXT** (§4.5), and in particular of the four bridges of Remark 4.5, which are the project's own reasoning and not certified by anything.
2. **Independent expert review of the overall proof.** Every mechanism this project has — reimplementation, certificates, Lean — is a mechanism against *implementation* error. None of them is a check on whether the argument is the right argument.
3. **Confirmation from one of the akr authors**, ideally: that "associated", as used in Observation 6 and Lemma 7, carries the cone dichotomy; and, separately, notice of the Observation 6 counterexample and of the two §10.2 counts that do not reproduce.
4. **The classification, E7/X1.** It is the largest unformalized object in the proof, it is the content of `hclass`, and it is the most plausible target for the next piece of formalization — unlike UB40, which is out of reach (§9).
5. **Conversion of this manifest into a readable paper.**
6. **The human author learning the argument well enough to defend it.**
7. Further Lean formalization *only* where it improves the trust boundary. In particular, chasing `hne40` by kernel computation is settled negative and should not be reattempted without a mathematical reduction of the domain first.

Formalizing all of Černý et al. or all of AKR is **not** required by this proof architecture. Those results are used as published black boxes, and the appropriate check on them is source auditing and expert review, not formalization.

# 13 Final trust statement

The project currently contains a heavily audited *candidate* computer-assisted proof of $f(7,7) = 38$. The novel finite reductions and the terminal elimination have been independently implemented, certificate checked, and in key places Lean-kernel verified. The principal remaining mathematical issue is A7-EXT: whether the published Observation 6 / Lemma 7 language licenses precisely the disjoint-status consequence the argument uses. That consequence has since been independently corroborated — by proof-checked unsatisfiability of an oriented-matroid encoding with a verified positive control, and by exact-arithmetic search — but its four bridges to the geometric statement remain the project's own reasoning, so it is recorded as VERY STRONGLY CORROBORATED, not verified. The same investigation found AKR's Observation 6 to be false as printed, a defect that does not reach this argument (§11, item 6). Conditional on A7-EXT, the deduction that the auxiliary graph $H$ is bipartite (A7-BRIDGE) is formally verified in Lean.

Two things should be said plainly alongside that. First, $f(7,7) = 38$ is **not** a Lean theorem and will not become one by further computation: the strongest formalised statement still carries `hclass` and `hne40`, and `hne40` is out of the kernel's reach by a factor of roughly 75 even after the fastest re-encoding measured (§9). Second, the mechanisms this project has used — independent reimplementation, certificates with standalone checkers, Lean-kernel verification — are all mechanisms against *implementation* error, and the record in §11 shows they work. None of them is a check on whether the argument is the right argument. Independent expert review is therefore the binding constraint, and it remains required before the result should be treated as established.

# References


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